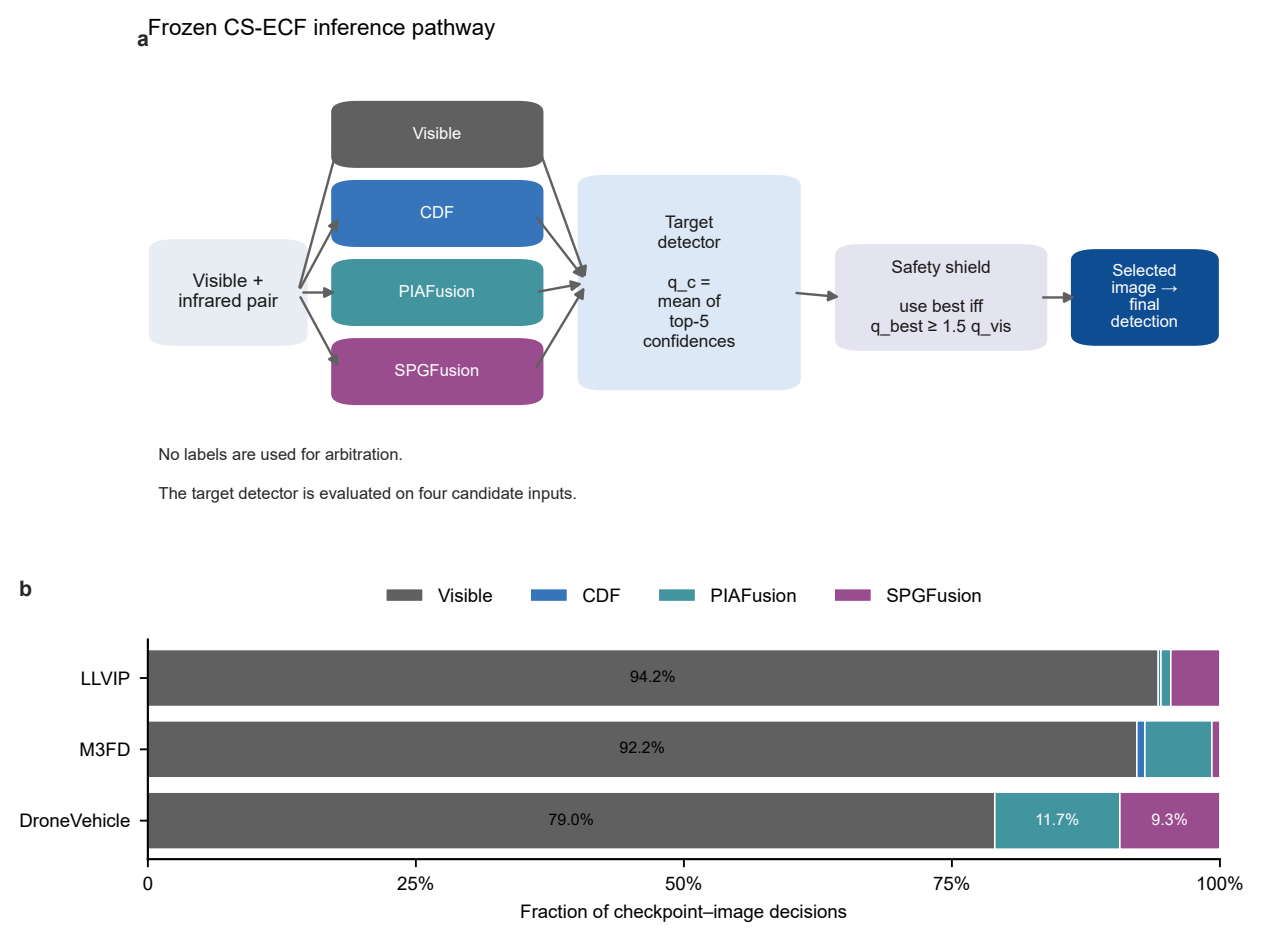
a
Frozen CS-ECF inference pathway
Visible
CDF
PIAFusion
SPGFusion
Visible + infrared pair
Target detector
q_c = mean of top-5 confidences
Safety shield
use best iff q_best ≥ 1.5 q_vis
Selected image → final detection
No labels are used for arbitration.
The target detector is evaluated on four candidate inputs.
b
Visible
CDF
PIAFusion
SPGFusion
LLVIP
M3FD
DroneVehicle
94.2%
92.2%
79.0%
11.7%
9.3%
0
25%
50%
75%
100%
Fraction of checkpoint–image decisions